Ex.No.2

23.7.26

# Study of Network Cables

AIM:

To understand different types of network cable.

1. Unshielded Twisted Pair (UTP):

Category 3 - 10 bps

Category 5 - Upto 100 Mbps

Category 5e - 1Gbps

Advantages:

* cheaper cost
* Easy to Install

Disadvantages:

more prone to (EMI) Electromagnetic interference

Application:

* 10 Base-T Ethernet
* Fast Ethernet, Gigabit Ethernet

## 2. Shielded Twisted Pair (STP):

i) Category 6, 6a - 10 Gbps

ii) Category 7 - 10 Gbps

Advantages:

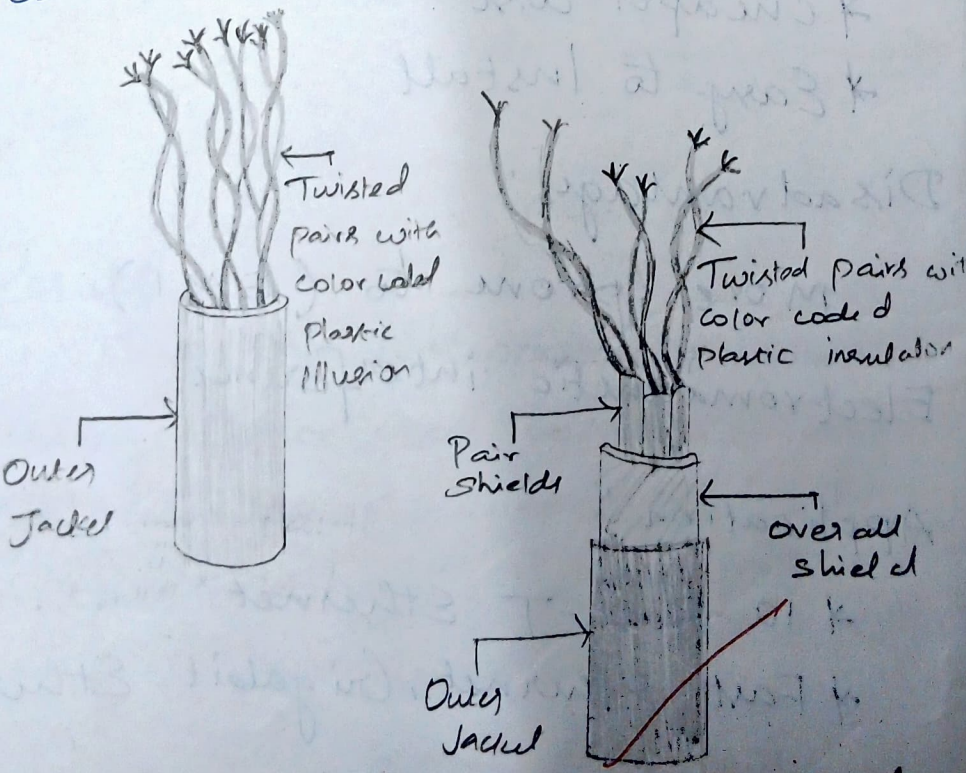
* Shielded, Faster than UTP

Disadvantage

* Expensive
* Greater installation effort

Application:

* Gigabit Ethernet, 10G Ethernet (55m) widely used in data Centres.




Shielded Twisted pair

3. Coaxial Cable

i) RG-6, RG-59, RG-11

10-100 Mbps

Advantages:

* High bandwidth
* Immune to Interference
* versatile

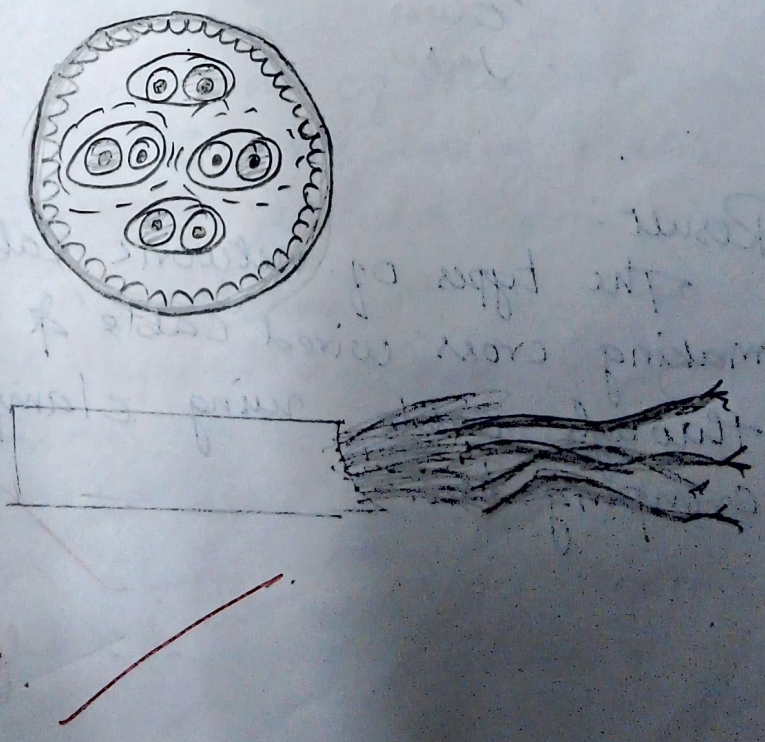
Disadvantages:

* Limited distance
* Cost
* Size is bulky.

Application:

Speed of Signal is 500 m

High speed internet Connections.



4. Fibre Optics Cable:

Single mode, Multi mode

100 Gbps

Advantages:

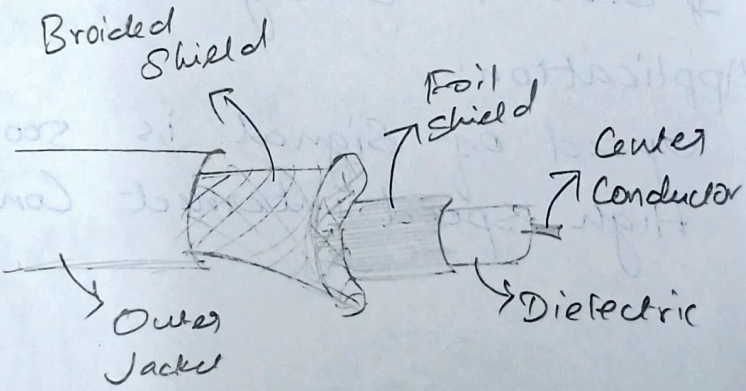
* High speed, bandwidth, security
* Long distance

Disadvantage:

* Expensive, Requires skilled installer

Application:

* Maximum distance of fibre optic cable is around 100 meters.




Result:

The types of network cables and making cross wired cable & Straight through cable using clamping/ crimping tool.